

Project ORION — Buyer Technical & Commercial FAQ

Authoritative Due Diligence Questions & Answers for CTOs, Quants & M&A Teams

REF: docs/acquisition/presentation/05-BUYER-FAQ.md
BASELINE COMMIT: d5908d0a0cc2feff99fa02573adb12b8eea33782
CLASSIFICATION: Confidential — Technical Due Diligence

1. PRODUCT & PLATFORM SCOPE

Q1: What is Project ORION?

Multi-tenant SaaS platform for quantitative FX research, algorithmic backtesting, optimization, walk-forward analysis, and simulated paper execution.

Q2: Is Project ORION a live trading engine?

No. The software operates strictly in paper simulation mode with **exactly \$0.00 live financial capital at risk**. Real-money clearing code is absent.

Q3: Is Project ORION a regulated platform?

No. ORION is strictly a software and intellectual property technology asset with no broker-dealer or RIA regulatory registrations.

2. ARCHITECTURE & TECHNOLOGY STACK

Q4: What is the architectural design pattern?

Hexagonal modular monorepo (ports and adapters). Pure business logic is isolated in **21 internal domain packages** with zero framework imports.

Q5: Why FastAPI and PostgreSQL 16?

FastAPI provides asynchronous ASGI performance with typed Pydantic v2 schemas. PostgreSQL 16 via async SQLAlchemy guarantees ACID multi-tenancy.

Q6: What happens if Redis is offline?

The system implements a graceful in-memory degraded fallback for sessions and rate limits without crashing the application.

3. QUANTITATIVE RESEARCH & BACKTESTING

Q7: How many algorithmic strategies exist?

4 concrete backtesting classes in StrategyRegistry (Trend Following, Mean Reversion, Breakout, Momentum) and 9 parameter schemas in API catalogue.

Q8: How is look-ahead bias prevented?

Sequential bar traversal protected by **LeakageGuard**, verifying temporal timestamp monotonicity at every indicator computation slice.

Q9: Are backtest results reproducible?

Yes. Identical inputs across identical bars produce mathematically identical trade journals and equity curves to the exact cent.

Q10: How is financial arithmetic handled?

Strictly uses Python Decimal arbitrary-precision math to avoid binary floating-point rounding errors.

4. OPTIMIZATION & WALK-FORWARD ANALYSIS

Q11: How does parameter optimization work?

Systematic multi-dimensional grid search across parameter bounds (POST /optimization/run) evaluating parameter sensitivity surfaces.

Q12: How does WFA detect curve-fitting?

Slices history into rolling In-Sample (IS) and Out-Of-Sample (OOS) windows; scores **Walk-Forward Efficiency (WFE)** to verify performance retention.

5. STRATEGY DEPLOYMENT LIFECYCLE

Q13: What is the deployment lifecycle?

Fail-closed state machine: PENDING_GATES → GATES_PASSED → INCUBATING → PAPER_VALIDATED → PROMOTION_CANDIDATE. Terminates at paper mode.

Q14: What quality gates are enforced?

Automated hurdles: WFE Threshold (≥ 0.50), Drawdown Ceiling ($\leq 15\%$), and Parameter Surface Stability before incubation entry.

6. PAPER TRADING & RISK MANAGEMENT

Q15: How does paper execution match orders?

PaperExecutionAdapter simulates realistic market spread, adverse slippage, latency, and netting. All orders stamped `is_paper=True`.

Q16: How are risk controls enforced?

Synchronous RiskService checks available margin, leverage ceilings (1:100 default), position sizes, and daily drawdown caps before order acceptance.

Q17: Is the autonomous worker active?

Source implemented with manual trigger (/worker/run-once); automated background loop is **disabled by default** (WORKER_ENABLED=false).

Project ORION — Buyer Technical & Commercial FAQ (Cont.)

Sections 7 through 9: Multi-Tenancy, Operations, and Acquisition Scope

REF: docs/acquisition/presentation/05-BUYER-FAQ.md
BASELINE COMMIT: d5908d0a0cc2feff99fa02573adb12b8eea33782
PAGE 2 OF 2

7. MULTI-TENANCY & SECURITY

Q18: How is tenant data isolated?

All queries bind `WHERE organization_id = :org_id` via dependency-injected `TenantContext`, mitigating cross-tenant data leakage.

Q19: What RBAC roles and permissions exist?

7 organization roles (Owner, Admin, PM, Risk Officer, Trader, Auditor, Viewer) governing **41 granular permissions** across 14 domains.

Q20: What is captured in the audit log?

`AuditLogModel` records operational actions (`ORDER_CREATE`, `USER_REGISTER`) with acting user IDs, timestamps, and structured JSON details.

8. DEPLOYMENT, OPERATIONS & DR

Q21: How is the platform deployed?

Declarative `render.yaml` PaaS blueprint orchestrating FastAPI web service, React static SPA, PostgreSQL 16, and Redis 7.

Q22: What is the baseline cloud hosting cost?

Baseline Render hosting configured for **~\$14/month** (\$7 DB + \$7 API under published 2026 pricing).

Q23: What evidence exists for disaster recovery?

In an isolated testing container, physical schema and data restoration across 29 tables was benchmarked in **~7.2 seconds** (Dossier 08).

9. COMMERCIAL STATUS & ACQUISITION SCOPE

Q24: What are the commercial revenue & customer metrics?

Revenue, paying customers, and active subscribers are **NOT ESTABLISHED IN REPOSITORY**. Sold strictly as a completed pre-revenue technology asset.

Q25: What is the authoritative subscription pricing model?

Implemented in source code as: **Free Sandbox (\$0)**, **Pro Trader (\$99)**, **Business Prop (\$299)**, and **Enterprise (Custom)**. Early \$49/\$149/\$399 proposals were superseded.

Q26: What does the buyer receive upon closing?

Proposed acquisition scope includes 100% transfer of the Git repository, 4,260 automated test baseline, database models, 15 migrations, and 18 closing due diligence dossiers, subject to executed transaction agreements. External vendor accounts (Render, Stripe, TwelveData) are independently provisioned by the buyer.